the low-entropy state of the early universe simply to be assumed; it feels hollow for time's arrow to be imposed on the universe, without explanation. At first blush, inflation offers progress by showing that what's assumed in the standard big bang emerges from inflationary evolution. But if the initiation of inflation requires yet other, highly special, exceedingly low-entropy conditions, we will pretty much find ourselves back at square one. We will merely have traded the big bang's special conditions for those necessary to ignite inflation, and the puzzle of time's arrow will remain just as puzzling.

What are the conditions necessary for inflation? We've seen that inflation is the inevitable result of the inflaton field's value getting stuck, for just a moment and within just a tiny region, on the high-energy plateau in its potential energy bowl. Our charge, therefore, is to determine how likely this starting configuration for inflation actually is. If initiating inflation proves easy, we'll be in great shape. But if the necessary conditions are extraordinarily unlikely to be attained, we will merely have shifted the question of time's arrow one step further back—to finding the explanation for the low-entropy inflaton field configuration that got the ball rolling.

I'll first describe current thinking on this issue in the most optimistic light, and then return to essential elements of the story that remain cloudy.

Boltzmann Redux

As mentioned in the previous chapter, the inflationary burst is best thought of as an event occurring in a preexisting universe, rather than being thought of as the creation of the universe itself. Although we don't have an unassailable understanding of what the universe was like during such a preinflationary era, let's see how far we can get if we assume that things were in a thoroughly ordinary, high-entropy state. Specifically, let's imagine that primordial, preinflationary space was riddled with warps and bumps, and that the inflaton field was also highly disordered, its value jumping to and fro like the frog in the hot metal bowl.

Now, just as you can expect that if you patiently play a fair slot machine, sooner or later the randomly spinning dials will land on triple